



Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

LAB REPORT

CSE3023 – WEB-BASED APPLICATION DEVELOPMENT

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Semester	2025/26-4
Lab Report	2
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Exercise

Objective: Develop a Product Inventory System by applying knowledge of Servlets, JavaBeans, and the DAO pattern to create a simple dynamic web application.

1. Database Setup: Create a new table named products in your CSE3023 database with the following columns:

- id (INT, Primary Key, Auto Increment)
- name (VARCHAR)
- category (VARCHAR)
- price (DOUBLE)
- quantity (INT)

QUERY
<pre>CREATE DATABASE CSE3023; USE CSE3023; CREATE TABLE products (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100), category VARCHAR(100), price DOUBLE, quantity INT);</pre>

2. Servlets (Controller): Create the necessary Servlets (e.g., InsertProductServlet, ViewProductServlet, UpdateProductServlet, DeleteProductServlet) to capture user inputs and interact with your ProductDAO. Strict rule: No JDBC code should be written inside these Servlets!

QUERY

com.lab.controller

```
1. DeleteProductServlet.java
/* Click
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e-default.txt to change this license
* Click
nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Ser
vlet.java to edit this template
*/
package com.lab.controller;
import com.lab.dao.ProductDAO;
import java.io.IOException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
/**
 *Name: Aina Nadhirah
 *Program: SMSK(SE)
 *Course: CSE3023
 *Lab: MP1
 *Date: 07/04/2026
 */
public class DeleteProductServlet extends HttpServlet {
    private ProductDAO dao;

    public void init() {
        dao = new ProductDAO();
    }

    protected void doGet(HttpServletRequest req,
        HttpServletResponse res)
        throws IOException {

        int id =
        Integer.parseInt(req.getParameter("id"));
        dao.deleteProduct(id);
    }
}
```

```

        res.sendRedirect("ViewProductServlet");
    }
}

```

InsertProductServlet.java

```

/*
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 */

package com.lab.controller;
import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
/**
 *Name: Aina Nadhirah
 *Program: SMSK(SE)
 *Course: CSE3023
 *Lab: MP1
 *Date: 07/04/2026
 */ public class InsertProductServlet extends HttpServlet
{
    private ProductDAO dao;

    @Override
    public void init() {
        dao = new ProductDAO();
    }

    @Override
    protected void doPost(HttpServletRequest req,
        HttpServletResponse res)
        throws ServletException, IOException {

        String name = req.getParameter("name");
        String category = req.getParameter("category");

```

```

        double price =
Double.parseDouble(req.getParameter("price"));
        int quantity =
Integer.parseInt(req.getParameter("quantity"));

        Product p = new Product(name, category, price,
quantity);
        dao.insertProduct(p);

        res.sendRedirect("ViewProductServlet");
    }
}

```

UpdateProductServlet.java

```

/*
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vlet.java to edit this template
*/
package com.lab.controller;
import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
/**
     *Name: Aina Nadhirah
     *Program: SMSK(SE)
     *Course: CSE3023
     *Lab: MP1
     *Date: 07/04/2026
*/
public class UpdateProductServlet extends HttpServlet {
    private ProductDAO productDAO;
    @Override
    public void init() {
        productDAO = new ProductDAO();
    }
}

```

```

@Override
protected void doGet(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    response.setContentType("text/html");
    PrintWriter out = response.getWriter();
    int id =
    Integer.parseInt(request.getParameter("id"));
    // Call DAO
    Product product = productDAO.selectProduct(id);
    if (product != null) {
        out.println("<h2>Update Product</h2>");
        out.println("<form
action='UpdateProductServlet' method='POST'>");
        out.println("<input type='hidden' name='id'
value='" + product.getId() + "'>");
        out.println("Name: <input type='text'
name='name' value='"
                        + product.getName() + "'
required><br><br>");
        out.println("Category: <input type='text'
name='category' value='"
                        + product.getCategory() + "'
required><br><br>");
        out.println("Price: <input type='number'
step='0.01' name='price' value='"
                        + product.getPrice() + "'
required><br><br>");
        out.println("Quantity: <input type='number'
name='quantity' value='"
                        + product.getQuantity() + "'
required><br><br>");
        out.println("<input type='submit'
value='Update Product'>");
        out.println("</form>");
    } else {
        out.println("<h3>Product not found!</h3>");
    }
}

@Override
protected void doPost(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    int id =

```

```

        Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category =
request.getParameter("category");
        double price =
Double.parseDouble(request.getParameter("price"));
        int quantity =
Integer.parseInt(request.getParameter("quantity"))
        Product product = new Product(id, name,
category, price, quantity);
        // Call DAO
        productDAO.updateProduct(product);

        response.sendRedirect("ViewProductServlet");
    }
}

```

ViewProductServlet.java

```

/*
 *Click
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nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Ser
vlet.java to edit this template
 */
package com.lab.controller;
import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
/**
 *Name: Aina Nadhirah
 *Program: SMSK(SE)
 *Course: CSE3023
 *Lab: MP1
 *Date: 07/04/2026
 */ public class ViewProductServlet extends HttpServlet {
    private ProductDAO dao;

```

```

@Override
public void init() {
    dao = new ProductDAO();
}

@Override
protected void doGet(HttpServletRequest req,
    HttpServletResponse res)
    throws IOException {

    res.setContentType("text/html");
    PrintWriter out = res.getWriter();

    List<Product> list = dao.selectAllProducts();

    out.println("<html>");
    out.println("<head>");
    out.println("<title>Product List</title>");
    out.println("</head>");
    out.println("<body>");

    out.println("<h2>Product List</h2>");
    out.println("<table border='1'>");

    out.println("<tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Qty</th><th>Action</th></tr>");

    for (Product p : list) {
        out.println("<tr>");
        out.println("<td>" + p.getId() + "</td>");
        out.println("<td>" + p.getName() +
"</td>");
        out.println("<td>" + p.getCategory() +
"</td>");
        out.println("<td>" + p.getPrice() +
"</td>");
        out.println("<td>" + p.getQuantity() +
"</td>");
        out.println("<td><a
href='UpdateProductServlet?id=" + p.getId() +
"'>Edit</a> | ");
        out.println("<a

```



```

        href='DeleteProductServlet?id=' + p.getId() +
        "'>Delete</a></td>");
        out.println("</tr>");
    }

    out.println("</table>");
    out.println("<br><br>");
    out.println("<a href='add_product.html'>+ Add
Product</a>");

    out.println("</body>");
    out.println("</html>");
}
}

```

3. Data Access Object (DAO): Create a ProductDAO class in the com.lab.dao package to handle all SQL operations (INSERT, SELECT, UPDATE, DELETE).

QUERY

```

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ava to edit this template

*/

package com.lab.dao;

/**
 *Name: Aina Nadhirah
 *Program: SMSK(SE)
 *Course: CSE3023
 *Lab: MP1

```

```

*Date: 07/04/2026

*/

import com.lab.model.Product;

import java.sql.*;

import java.util.ArrayList;

import java.util.List;

public class ProductDAO {

    private String jdbcURL =
        "jdbc:mysql://localhost:3306/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";

    protected Connection getConnection() {
        Connection conn = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            conn = DriverManager.getConnection(jdbcURL,
jdbcUsername, jdbcPassword);
        } catch (Exception e) {
            e.printStackTrace();
        }
        return conn;
    }

    // CREATE
    public void insertProduct(Product product) {
        String sql = "INSERT INTO products (name,
category, price, quantity) VALUES (?, ?, ?, ?)";
        try (Connection conn = getConnection());

```

```

        PreparedStatement ps =
conn.prepareStatement(sql)) {

        ps.setString(1, product.getName());
        ps.setString(2, product.getCategory());
        ps.setDouble(3, product.getPrice());
        ps.setInt(4, product.getQuantity());

        ps.executeUpdate();
    } catch (Exception e) {
        e.printStackTrace();
    }
}

// READ ALL
public List<Product> selectAllProducts() {
    List<Product> list = new ArrayList<>();
    String sql = "SELECT * FROM products";

    try (Connection conn = getConnection();
        PreparedStatement ps =
conn.prepareStatement(sql)) {

        ResultSet rs = ps.executeQuery();

        while (rs.next()) {
            list.add(new Product(
                rs.getInt("id"),
                rs.getString("name"),
                rs.getString("category"),
                rs.getDouble("price"),
                rs.getInt("quantity")
            ));
        }
    }
}

```

```

    }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}

// READ ONE
public Product selectProduct(int id) {
    Product p = null;
    String sql = "SELECT * FROM products WHERE
id=?";

    try (Connection conn = getConnection();
        PreparedStatement ps =
conn.prepareStatement(sql)) {

        ps.setInt(1, id);
        ResultSet rs = ps.executeQuery();

        if (rs.next()) {
            p = new Product(
                id,
                rs.getString("name"),
                rs.getString("category"),
                rs.getDouble("price"),
                rs.getInt("quantity")
            );
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return p;
}

```

```

    }

    // UPDATE
    public boolean updateProduct(Product p) {
        String sql = "UPDATE products SET name=?,
category=?, price=?, quantity=? WHERE id=?";
        boolean updated = false;

        try (Connection conn = getConnection();
            PreparedStatement ps =
conn.prepareStatement(sql)) {

            ps.setString(1, p.getName());
            ps.setString(2, p.getCategory());
            ps.setDouble(3, p.getPrice());
            ps.setInt(4, p.getQuantity());
            ps.setInt(5, p.getId());

            updated = ps.executeUpdate() > 0;
        } catch (Exception e) {
            e.printStackTrace();
        }
        return updated;
    }

    // DELETE
    public boolean deleteProduct(int id) {
        String sql = "DELETE FROM products WHERE id=?";
        boolean deleted = false;

        try (Connection conn = getConnection();
            PreparedStatement ps =
conn.prepareStatement(sql)) {

```

```

        ps.setInt(1, id);
        deleted = ps.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
    }
    return deleted;
}
}

```

UserDAO.java

```

/*
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 ava to edit this template
 */

package com.lab.dao;

/**
 *Name: Aina Nadhirah
 *Program: SMSK(SE)
 *Course: CSE3023
 *Lab: MP1
 *Date: 07/04/2026
 */

//class userDAO

import com.lab.model.User;

import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.PreparedStatement;

```

```

import java.sql.ResultSet;

import java.sql.SQLException;

import java.util.ArrayList;

import java.util.List;

public class UserDao {

    private String jdbcURL =
        "jdbc:mysql://localhost:3306/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";

    // Method untuk mendapatkan sambungan Database
    protected Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection =
                DriverManager.getConnection(jdbcURL, jdbcUsername,
                    jdbcPassword);
        }
        catch (SQLException | ClassNotFoundException e)
        {
            e.printStackTrace();
        }

        return connection;
    }

    // CREATE: Masukkan pengguna baru
    public void insertUser(User user) {
        String sql = "INSERT INTO users (username,
            password, roles)VALUES (?, ?, ?)";
        try (Connection conn = getConnection();
            PreparedStatement pstmt =
                conn.prepareStatement(sql)) {

            pstmt.setString(1, user.getUsername());
            pstmt.setString(2, user.getPassword());
            pstmt.setString(3, user.getRoles());

            pstmt.executeUpdate();
        }
        catch (SQLException e)
        {
            e.printStackTrace();
        }
    }
}

```

```

    }
}

// READ: Ambil satu pengguna berdasarkan ID (Untuk
Update)
public User selectUser(int id) {
    User user = null;
    String sql = "SELECT id, username, password,
roles FROM users WHERE id =?";

    try (Connection conn = getConnection();
        PreparedStatement pstmt =
conn.prepareStatement(sql)) {

        pstmt.setInt(1, id);
        ResultSet rs = pstmt.executeQuery();

        if (rs.next()) {

            String username =
rs.getString("username");
            String password =
rs.getString("password");
            String roles = rs.getString("roles");

            user = new User(id, username, password,
roles);
        }

    } catch (SQLException e) {
        e.printStackTrace();
    }
    return user;
}

// READ: Ambil senarai semua pengguna (Untuk View)
public List<User> selectAllUsers() {
    List<User> users = new ArrayList<>();
    String sql = "SELECT * FROM users";

    try (Connection conn = getConnection();
        PreparedStatement pstmt =
conn.prepareStatement(sql)) {

        ResultSet rs = pstmt.executeQuery();

        while (rs.next()) {
            int id = rs.getInt("id");

```



```

        String username =
rs.getString("username");
        String password =
rs.getString("password");
        String roles =
rs.getString("roles");
        users.add(new User(id, username,
password, roles));
    }
    } catch (SQLException e)
    {
        e.printStackTrace();
    }
    return users;
}

// UPDATE: Kemaskini maklumat pengguna
public boolean updateUser(User user) {
    boolean rowUpdated = false;
    String sql = "UPDATE users SET username = ?,
password = ?, roles = ? WHERE id = ?";

    try (Connection conn = getConnection();
        PreparedStatement pstmt =
conn.prepareStatement(sql)) {

        pstmt.setString(1, user.getUsername());
        pstmt.setString(2, user.getPassword());
        pstmt.setString(3, user.getRoles());
        pstmt.setInt(4, user.getId());

        rowUpdated = pstmt.executeUpdate() > 0;

    } catch (SQLException e) {
        e.printStackTrace();
    }
    return rowUpdated;
}

// DELETE: Padam pengguna
public boolean deleteUser(int id) {
    boolean rowDeleted = false;
    String sql = "DELETE FROM users WHERE id = ?";

    try (Connection conn = getConnection();
        PreparedStatement pstmt =
conn.prepareStatement(sql)) {

        pstmt.setInt(1, id);

```

```

        rowDeleted = pstmt.executeUpdate() > 0;

    } catch (SQLException e) {
        e.printStackTrace();
    }
    return rowDeleted;
}
}

```

4. JavaBean (Model): Create a class named Product in the com.lab.model package. Ensure it has the appropriate private attributes, constructors (empty and parameterized), and getter/setter methods.

QUERY
Product.java <pre> /* *Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license- default.txt to change this license *Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template */ package com.lab.model; /** *Name: Aina Nadhirah *Program: SMSK(SE) *Course: CSE3023 *Lab: MP1 *Date: 07/04/2026 </pre>

```

*/

public class Product {

private int id; private String name; private String
category; private double price; private int quantity;

// Empty constructor
public Product() {}

// Constructor without ID (Insert)
public Product(String name, String category, double price,
int quantity) {
    this.name = name;
    this.category = category;
    this.price = price;
    this.quantity = quantity;
}

// Constructor with ID (Update)
public Product(int id, String name, String category, double
price, int quantity) {
    this.id = id;
    this.name = name;
    this.category = category;
    this.price = price;
    this.quantity = quantity;
}

// Getter & Setter
public int getId() { return id; }
public void setId(int id) { this.id = id; }

public String getName() { return name; }

```

```

public void setName(String name) { this.name = name; }

public String getCategory() { return category; }
public void setCategory(String category) { this.category =
category; }

public double getPrice() { return price; }
public void setPrice(double price) { this.price = price; }

public int getQuantity() { return quantity; }
public void setQuantity(int quantity) { this.quantity =
quantity; }

}

```

User.java

```

/*
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 default.txt to change this license
 *Click
 nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java
 to edit this template

 */ package com.lab.model;

/**
 *Name: Aina Nadhirah
 *Program: SMSK(SE)
 *Course: CSE3023
 *Lab: MP1
 *Date: 07/04/2026
 */

public class User {
private int id;

```

```

private String username;

private String password;

private String roles;

// Constructor Kosong (Wajib untuk JavaBean)
public User() {

}

// Constructor dengan ID (Untuk Update & Delete)
public User(int id, String username, String password,
String roles) {
    this.id = id;
    this.username = username;
    this.password = password;
    this.roles = roles;
}

// Constructor tanpa ID (Untuk Insert)
public User(String username, String password, String roles)
{
    this.username = username;
    this.password = password;
    this.roles = roles;
}

// Getter dan Setter
public int getId() {
    return id;
}

public void setId(int id) {
    this.id = id;
}

public String getUsername() {
    return username;
}

public void setUsername(String username) {
    this.username = username;
}

public String getPassword() {
    return password;
}

```

```

public void setPassword(String password) {
    this.password = password;
}

public String getRoles() {
    return roles;
}

public void setRoles(String roles) {
    this.roles = roles;
}

}

```

5. User Interface (View): Create an HTML form (add_product.html) to insert new products. Ensure your view servlet generates an HTML table displaying all products, along with functional "Edit" and "Delete" hyperlinks for each row.

QUERY
add_product.html <pre> <!DOCTYPE html> <!-- Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license- default.txt to change this license Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Html.html to edit this template --> <!-- Name: Aina Nadhirah Program: SMSK(SE) Course: CSE3023 Lab: MP1 Date: 07/04/2026 --> </pre>

```

<html>
<head>
    <title>Add User</title>
</head>
<body>
    <h2>Add New Product</h2>
    <form action="InsertProductServlet" method="POST">
        Name: <input type="text" name="name" required><br><br>
        Category: <input type="text" name="category"
required><br><br>
        Price: <input type="number" step="0.01" name="price"
required><br><br>
        Quantity: <input type="number" name="quantity"
required><br><br>
        <input type="submit" value="Add Product">
    </form>

    <br>
    <a href="ViewProductServlet">View Product</a>
</body>
</html>

```

Index.html

```

<!DOCTYPE html>
<!--
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to edit this template
-->

<!--

```

Name: Aina Nadhirah

Program: SMSK(SE)

Course: CSE3023

Lab: MP1

Date: 07/04/2026

-->

```
<html>
```

```
<head>
```

```
    <title>Add User</title>
```

```
</head>
```

```
<body>
```

```
    <h2>Add New User</h2>
```

```
    <form action="InsertServlet" method="POST">
```

```
        Username: <input type="text" name="username"
required><br><br>
```

```
        Password: <input type="password" name="password"
required><br><br>
```

```
        Role:
```

```
        <select name="roles">
```

```
            <option value="Admin">Admin</option>
```

```
            <option value="Staff">Staff</option>
```

```
            <option value="Student">Student</option>
```

```
        </select><br><br>
```

```
        <input type="submit" value="Submit">
```

```
    </form>
```

```
    <br>
```

```
    <a href="ViewServlet">View Users</a>
```

```
</body>
```

```
</html>
```


Output:

Enter product:

Add New Product

Name:

Category:

Price:

Quantity:

[View Product](#)

View product:

Product List

ID	Name	Category	Price	Qty	Action
1	Aburi Salmon	Sushi	3.5	35	Edit Delete

[+ Add Product](#)

Add new product:

Add New Product

Name:

Category:

Price:

Quantity:

[View Product](#)

Product List

ID	Name	Category	Price	Qty	Action
1	Aburi Salmon	Sushi	3.5	35	Edit Delete
3	Unagi	Sushi	3.2	40	Edit Delete

[+ Add Product](#)

Update product:

Update Product

Name:

Category:

Price:

Quantity:

Product List

ID	Name	Category	Price	Qty	Action
1	Aburi Salmon	Sushi	3.5	35	Edit Delete
3	Unagi	Sushi	3.2	67	Edit Delete

[+ Add Product](#)

Delete product

Product List

ID	Name	Category	Price	Qty	Action
3	Unagi	Sushi	3.2	67	Edit Delete

[+ Add Product](#)

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

[SELECT](#) * FROM `products`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

[+T-](#) [id](#) [name](#) [category](#) [price](#) [quantity](#)

☐ [Edit](#) [Copy](#) [Delete](#) 3 Unagi Sushi 3.2 67

[↑](#) ☐ Check all With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: